RESEARCH OBJECTIVE 2 · DATASET PAPER

Portfolio Drawdown Risk Labeling of Nifty 50 Sector Portfolios During Macro Stress Events

COVID-19 · RBI Rate Hikes · Russia-Ukraine · US Fed Tightening · 2020–2023

1. Research Overview

This objective proposes the construction of **NiftyStress-44**, the first publicly available benchmark dataset capturing sector-level portfolio drawdown behavior during four identified macro stress events in Indian markets (2020–2023). Existing Indian market datasets cover normal trading periods only. No dataset isolates and labels portfolio-level drawdown metrics (MDD, VaR, Sharpe drop, recovery time) specifically during crisis windows — making it impossible to train or benchmark portfolio risk classifiers on crisis behavior. This paper fills that gap with a fully reproducible pipeline using only public OHLCV data and standard Python libraries.

Dataset Paper | Portfolio Risk | 4 Stress Events | 11 Sectors | MDD + VaR | Risk Labels

2. Research Gap

- Existing Indian market datasets (BSE/NSE OHLCV on Kaggle, Mendeley) cover normal trading — none isolate crisis windows.
- Portfolio risk models trained on normal-period data fail during stress events because crisis behavior is structurally different.
- No dataset provides a sector x stress-event risk label matrix for NSE — forcing researchers to build it from scratch each time.
- MDD, VaR, and recovery time are rarely combined in a single labeled dataset for Indian emerging market sectors.

3. Objective Statement

To construct a labeled portfolio risk benchmark by identifying four macro stress windows in Indian markets (2020–2023), computing sector-level portfolio risk metrics (MDD, VaR at 95%, recovery days, Sharpe ratio drop) for all 11 NSE sectors within each window, and assigning a three-class risk label (High / Medium / Low) based on MDD thresholds — producing a dataset directly usable for training and benchmarking portfolio risk classifiers.

4. Defined Macro Stress Windows

Stress Event	Date Window	Trigger	Expected Impact
COVID-19 Crash	Feb 2020 – Apr 2020	WHO pandemic declaration	High MDD across all sectors
Russia-Ukraine Shock	Feb 2022 – Apr 2022	Invasion announcement (Feb 24, 2022)	Energy & Metals: Up; IT: Down
RBI Rate Hike Cycle	May 2022 – Feb 2023	RBI repo rate 4.0% → 6.5%	Banking volatility, FMCG stable
US Fed Tightening	Mar 2022 – Dec 2023	Fed Funds rate 0.25% → 5.5%	FII outflows, IT sector stress

5. Risk Label Definition

Risk Label	MDD Threshold	Interpretation
High Risk	MDD < -15%	Severe capital drawdown; recovery likely > 60 days
Medium Risk	-15% <= MDD < -5%	Moderate loss; partial hedging would have helped
Low Risk	MDD >= -5%	Resilient sector; minimal portfolio impact

6. Step-by-Step Implementation Flow

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Step 1: Define and Validate Stress Windows

• Identify the 4 macro stress events with precise start and end dates based on financial literature and RBI/NSE press releases.

• Validate window boundaries: confirm each window captures a peak-to-trough drawdown of at least 5% in the Nifty 50 index.

• Document trigger event for each window (e.g., WHO pandemic declaration for COVID, RBI MPC meeting dates for rate hikes).

• Store window metadata as a configuration file used throughout the pipeline.

Tools: *pandas, yfinance (for index-level validation)*

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Step 2: Fetch Sector-Level OHLCV Data

• For each of the 11 NSE sectors, identify the constituent ETF or compute an equal-weighted sector index using stock tickers.

• Fetch daily OHLCV data for each sector portfolio from yfinance for the full 2019–2024 period (to include pre-stress baseline).

• Compute daily portfolio returns: equal-weighted average of constituent stock daily returns.

• Handle missing data: forward-fill up to 2 consecutive missing trading days; flag windows with > 2 missing days.

Tools: *yfinance, pandas, numpy*

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Step 3: Compute Maximum Drawdown (MDD)

• For each sector x stress window combination, compute the running maximum of cumulative returns within the window.

• MDD = (Trough Value - Peak Value) / Peak Value x 100.

• Record: peak date, trough date, MDD percentage, and number of days from peak to trough.

• Cross-validate with published NSE index drawdown figures for the COVID window as a sanity check.

Tools: *numpy, pandas*

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Step 4: Compute VaR, Sharpe Drop, and Recovery Time

• Value at Risk (VaR at 95%): compute the 5th percentile of daily returns within the stress window using historical simulation.

• Sharpe Drop: Sharpe ratio (annualised) during stress window minus Sharpe ratio during the 90-day pre-stress baseline period.

• Recovery Days: number of trading days from the trough date until the sector portfolio price returns to its pre-stress peak level.

• If a sector has not recovered by the window end date, flag recovery_days as NaN and note in data dictionary.

Tools: *numpy, pandas, scipy*

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Step 5: Assign Risk Labels and Validate

• Apply MDD threshold rules: High (MDD < -15%), Medium (-15% to -5%), Low (MDD >= -5%).

• Produce the primary output: an 11 x 4 risk label matrix (sectors x stress events) with all computed metrics.

• Run cross-validation: compare risk labels against domain knowledge (e.g., Banking should be High during COVID, Gold/FMCG Low).

• Spot-check 10 sector-event combinations manually against public NSE data to validate metric computation.

Tools: *pandas, Excel for validation*

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Step 6: Baseline Classifier and Release

• Train a simple Random Forest classifier (scikit-learn) on the 44-record dataset to predict risk label from metrics as a baseline.

• Report 3-fold cross-validation accuracy, precision, recall per class as benchmark numbers for future models to beat.

• Export dataset as CSV + Excel with full data dictionary; upload to Zenodo with DOI.

• Publish scraping and computation code as open-source GitHub repository for reproducibility.

Tools: *scikit-learn, pandas, Zenodo, GitHub*

7. Output Dataset Schema

Column	Type	Description / Example
sector	TEXT	Banking
stress_event	TEXT	COVID-19 Crash
window_start	DATE	2020-02-17
window_end	DATE	2020-04-01
max_drawdown_pct	FLOAT	-38.4 (peak-to-trough %)
var_95	FLOAT	-4.2 (daily VaR at 95%)
recovery_days	INT	87 (days to recover to pre-stress level)
sharpe_drop	FLOAT	-1.35 (Sharpe ratio change during window)
risk_label	TEXT	High / Medium / Low

8. Success / Evaluation Metrics

Metric	Target
Stress event windows covered	4 (COVID, Russia-Ukraine, RBI, US Fed)
Sectors covered	11 NSE sectors
Final labeled records	44 records (11 sectors x 4 events)
Risk metric coverage	MDD, VaR (95%), Recovery Days, Sharpe Drop
Baseline classifier accuracy	>= 80% (Random Forest 3-class)
Data completeness	>= 98% (yfinance coverage)

9. Tools, Libraries & Timeline

Task	Tool / Library	Week
Stress window definition & validation	pandas, yfinance, NSE data	1
Sector OHLCV fetch & return computation	yfinance, pandas, numpy	1–2
MDD computation	numpy, pandas	2
VaR, Sharpe Drop, Recovery Days	numpy, pandas, scipy	2–3
Risk label assignment & validation	pandas, Excel spot-check	3
Baseline Random Forest classifier	scikit-learn	4
Documentation & Zenodo release	Zenodo, GitHub, Markdown	5–6

10. Target Conference Venues

[Primary] IEEE CiFer 2025 — Computational Intelligence for Financial Engineering

[Primary] Springer FinanceML — Emerging Markets Risk Benchmark Track

[Secondary] ICACCS 2025 — Indian Market Portfolio Analysis

[Secondary] ACM ICAIF 2025 — Dataset & Benchmark Paper Track

11. Files to be Submitted

All files must be submitted as a single zipped folder named **StudentName_Objective2/** containing the following:

Category	File Name	Description
Raw Data	**sector_returns/**	Folder of per-sector daily return series CSVs for full period 2019–2024
Raw Data	**stress_windows_config.json**	Defined start/end dates for all 4 stress events with trigger notes
Final Dataset	**NiftyStress44_final.csv**	Complete labeled dataset with all 9 schema columns
Final Dataset	**risk_label_matrix.xlsx**	11 sectors x 4 stress events grid showing risk label at a glance
Final Dataset	**data_dictionary.xlsx**	Column definitions, MDD thresholds, and label rules
Validation	**baseline_classifier_results.csv**	Random Forest 3-fold CV accuracy, precision, recall per class
Validation	**domain_validation_notes.xlsx**	Spot-check comparing computed labels against known financial outcomes
Code	**sector_ohlcv_fetch.py**	Sector return computation from constituent tickers
Code	**mdd_compute.py**	Maximum Drawdown calculation per stress window
Code	**risk_metrics.py**	VaR, Sharpe Drop, and Recovery Days computation
Code	**label_assign.py**	MDD threshold to risk label assignment
Code	**baseline_classifier.py**	Random Forest training and evaluation script
Code	**requirements.txt**	Python library names and versions used

12. How the Supervisor Uses These Files

File	Used For
NiftyStress44_final.csv	Feed into parent system backtesting node (O3) as stress-period risk evaluation benchmark
risk_label_matrix.xlsx	Reference table for portfolio risk discussion section in research paper
data_dictionary.xlsx	Cite in methodology section of research paper for data provenance
baseline_classifier_results.csv	Report baseline accuracy figure in dataset paper as benchmark for future models
domain_validation_notes.xlsx	Evidence of label validity — include in paper's data quality section
requirements.txt	Include in reproducibility appendix of the paper